

Amaan A. Javed

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Education

Queen's University

Bachelor of Computer Science (Honours), Specialization in Software Design

Expected May 2027

Kingston, Canada

- GPA: 3.95/4.00
- **Awards:** Mitacs Globalink Research Award (\$6,000 research grant), E.M.S. Technologies Scholarship (Top Software Design Student), Honour Roll with Distinction (4x), Dean's Award of Excellence, Principal's Scholarship, M. C. Urquhart Book Prize in Economics

Work Experience

Mission Control Space Services

Full-Stack Software Engineering Intern

Sep 2025 – Aug 2026

Ottawa, Canada

- Built a containerized Python interface that connected a rover operations platform to a lunar simulator, enabling **40+ developers and external stakeholders** to run simulated rover missions.
- Architected and led the implementation of a **role-based access control** system across **Vue.js** and **Django**, replacing global permissions with granular, resource-level authorization.
- Improved mission-planning canvas latency by **43%** by implementing **WebSocket** updates and **AWS S3** storage.
- Reduced application crashes by **40%** by eliminating race conditions between **Vuex** and **Redis** during concurrent layout updates.
- Automated regression testing with **Playwright** and **GitLab CI**, validating end-to-end user workflows on every merge request.

Tangerine Bank

DevOps Engineering Intern

May 2024 – Aug 2024

Toronto, Canada

- Reduced CI build failures by **40%** by developing shell tooling to analyze Docker failures and resolve Docker image defects.
- Created an on-demand Jenkins job with Groovy and the GitHub API to automate release PR aggregation, saving **30+ min/release**.

Scotiabank

Software Developer Intern

May 2023 – Aug 2023

Toronto, Canada

- Built a Python ticket search tool for **1,600+ support tickets**, achieving **80% retrieval accuracy** and reducing issue resolution time.
- Reduced runtime of scheduled financial data extraction jobs by up to **35%** by optimizing SQL Server and Oracle SQL queries.

Research Experience

Perk Lab - Laboratory for Percutaneous Surgery

Undergraduate Research Student

Sep 2026 – Present

Kingston, Canada

- Developing the **Lumbar Tutor**, a lumbar puncture training simulator that tracks needle insertion to guide and assess trainees.
- Building object detection and workflow recognition (CNN-LSTM) models that track the syringe and recognize procedural steps, enabling assessment from a low-cost camera.

Image Guided Therapy Research Group (IGT)

Visiting Undergraduate Research Student

May 2025 – Aug 2025

Madrid, Spain

- Developed an **augmented reality** laparoscopic training simulator using HoloLens 2 and 3D-printed anatomical models.
- Applied **deep learning** for camera parameter estimation and 3D reconstruction, generating anatomical models from RGB images.

Projects

[Coursify](#) | Next.js, TypeScript, Supabase, Redis, LangChain, pgvector

- Founded and led a team of **8 developers** to build a course planning platform used by **3,000+ Queen's students**.
- Built an AI agent using LangChain, RAG, and 15+ tools to retrieve and reason over grades, reviews, and degree requirements.

[RateMyProfessors API Client](#) | Python, TypeScript, GraphQL

- Reverse-engineered RateMyProfessors GraphQL API and published clients on PyPI and npm with **5,600+ total downloads**.

[OmniLRS \(Open Source Lunar Simulator\)](#) | NVIDIA Isaac Sim, Python, YAMCS

- Implemented a YAMCS controller for the Clearpath Husky rover, supporting telemetry, commands, and live camera streaming.

Technical Skills

Languages:	Python, Java, JavaScript, TypeScript, C, Bash, SQL, HTML/CSS
Frameworks & DB:	Flask, Django, Vue.js, React.js, Node.js, Next.js, GraphQL, Prisma, Supabase, PostgreSQL, MongoDB
ML & Data:	PyTorch, TensorFlow, Hugging Face, LangChain, Pandas, NumPy, pgvector
Cloud & Infra:	AWS (ECS, ALB, S3, IAM), Terraform, Docker, Jenkins, Redis, Grafana, Heroku, Vercel, Git